

Factors Associated with Life Satisfaction Among Women in Canada

An Analysis of Socioeconomic, Employment, and Health-Related
Determinants

Using Canadian Community Health Survey (CCHS 2022) Data

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Abstract

Life satisfaction is an important indicator of subjective well-being and reflects how individuals evaluate their overall quality of life. Understanding the factors associated with life satisfaction is particularly important for women, who may experience unique socioeconomic, employment-related, and health-related challenges that influence their well-being. This study examines factors associated with life satisfaction among women in Canada using data from the 2022 Canadian Community Health Survey (CCHS).

The study focuses on demographic, socioeconomic, employment-related, and health-related determinants of life satisfaction. After selecting female respondents and applying data cleaning procedures, an exploratory analysis dataset consisting of 52,884 women was created. Descriptive statistics and exploratory data analysis were conducted to examine the distribution of life satisfaction and investigate relationships between life satisfaction and selected explanatory variables, including age, education, household income, food security, employment status, functional difficulty, chronic health conditions, and province of residence.

The exploratory findings indicate that socioeconomic and health-related factors are strongly associated with life satisfaction. Women with higher household incomes generally reported higher levels of life satisfaction, while women experiencing food insecurity reported substantially lower levels of life satisfaction. Functional difficulties and chronic health conditions were also associated with lower life satisfaction. Employment status

showed meaningful differences in well-being, suggesting that labour force participation and employment conditions may influence life satisfaction through both economic and social pathways.

Building on these findings, the study proposes the use of interpretable predictive modeling approaches, including Logistic Regression and Decision Tree Classification, to identify factors associated with lower life satisfaction among women. The results of this research are expected to contribute to a better understanding of women’s well-being in Canada and provide insights into the socioeconomic, employment-related, and health-related conditions that may influence subjective well-being in the post-pandemic context.

1 Introduction

Life satisfaction is widely recognized as an important indicator of subjective well-being and overall quality of life. Unlike clinical measures of mental illness, life satisfaction captures how individuals evaluate their lives as a whole and reflects broader social, economic, health-related, and personal circumstances. As a result, life satisfaction has become an increasingly important outcome in public health, social science, and population health research.

Women’s well-being is influenced by a complex combination of demographic, socioeconomic, employment-related, and health-related factors. Although women have made significant gains in education and labour force participation over recent decades, many continue to experience unique challenges related to income inequality, caregiving responsibilities, work-family balance, employment conditions, and health-related limitations. These factors may influence how women perceive their quality of life and overall satisfaction with life.

Previous research has shown that socioeconomic conditions play an important role in shaping subjective well-being. Income, educational attainment, food security, and employment status have consistently been associated with differences in life satisfaction across populations. Individuals with greater financial security and access to resources generally report higher levels of life satisfaction, while those experiencing economic hardship, unemployment, or material deprivation often report lower levels of well-being.

Health-related factors are also important determinants of life satisfaction. Functional limitations, chronic health conditions, and difficulties performing everyday activities may affect independence, employment participation, social engagement, and quality of life. Consequently, health and well-being are closely interconnected, particularly among populations facing multiple social and economic pressures.

The COVID-19 pandemic further highlighted the importance of understanding factors associated with well-being. Employment disruptions, rising living costs, financial uncertainty, caregiving demands, and changing social conditions have affected many Canadians and may continue to influence life satisfaction in the post-pandemic period. Examining these factors using recent national survey data can provide valuable insights into the determinants of well-being among women in Canada.

Although previous studies have examined individual determinants of well-being, many have focused on specific factors such as income, employment, or health in isolation. Fewer studies have explored how demographic, socioeconomic, employment-related, and health-related factors jointly influence life satisfaction among women using recent Canadian population-level data. Furthermore, there remains a need for interpretable analytical approaches that can identify the factors most strongly associated with lower life satisfaction.

This study uses data from the 2022 Canadian Community Health Survey (CCHS) to examine factors associated with life satisfaction among women in Canada. Specifically, the study investigates how demographic characteristics, socioeconomic conditions, employment-related factors, and health-related variables are associated with life satisfaction and subjective well-being. By combining exploratory analysis with interpretable predictive modeling techniques, the study aims to identify important determinants of life satisfaction and contribute to a better understanding of women’s well-being in the Canadian context.

2 Literature Review

2.1 Life Satisfaction and Subjective Well-Being

Life satisfaction is an important component of subjective well-being and represents an individual’s cognitive evaluation of their overall quality of life. Unlike clinical mental health measures, life satisfaction does not diagnose depression, anxiety, or psychological disorder. Instead, it captures how individuals evaluate their lives as a whole, making it a useful outcome for population-level studies of well-being.

Diener (1984) defined subjective well-being as a broad concept that includes life satisfaction, positive affect, and negative affect. This framework emphasizes that well-being is not limited to the absence of illness, but also includes individuals’ evaluations of their lives and emotional experiences. Life satisfaction is therefore widely used in social science and public health research as an indicator of subjective well-being and quality of life.

This distinction is important for the current study because the Canadian Community Health

Survey 2022 does not provide a suitable clinical mental health outcome for the full analytical sample. However, it includes a grouped life satisfaction variable, which provides a meaningful measure of subjective well-being. Using life satisfaction as the outcome allows this study to examine how socioeconomic, employment-related, demographic, and health-related factors are associated with well-being among women in Canada.

Research on subjective well-being has shown that life satisfaction is influenced by multiple factors, including income, employment, health, social support, and personal circumstances. Diener (1984) argued that well-being should be understood as a multidimensional construct shaped by both individual and social conditions. Therefore, examining life satisfaction among women requires attention to broader determinants such as financial security, food security, employment participation, functional health, and social context.

2.2 Women and Life Satisfaction

Women’s life satisfaction is shaped by a combination of social, economic, health-related, and gender-related factors. Previous research has shown that women may experience unique pressures related to employment, caregiving, household responsibilities, income inequality, and social expectations. These pressures can influence overall well-being and quality of life.

Denton et al. (2004) examined gender differences in health in Canada and found that health outcomes are strongly influenced by psychosocial, structural, and behavioural determinants. Their study showed that women’s health and well-being are affected by employment conditions, income, caregiving responsibilities, and chronic stress exposure. These findings are relevant to the current study because they support the idea that women’s well-being should be examined within a broader social determinants framework rather than through individual characteristics alone.

Gender roles and social relationships may also influence life satisfaction. Matud et al. (2014) examined life satisfaction among adult women and men and found that self-esteem and social support were among the strongest predictors of life satisfaction. Their study also showed that social support played an important role in women’s life satisfaction. This suggests that women’s subjective well-being may be shaped not only by economic and health conditions, but also by social relationships and psychosocial resources.

Research on work-family dynamics further supports the importance of gendered experiences in well-being. Bilodeau et al. (2025) found that women often experience greater work-family conflict and emotional strain due to unequal caregiving and domestic responsibilities. Similarly, Artazcoz et al. (2004) showed that combining paid work and family demands

affects men and women differently, with women often experiencing greater health-related consequences when managing both employment and caregiving responsibilities.

Overall, existing literature suggests that women’s life satisfaction is influenced by multiple overlapping factors, including socioeconomic status, employment conditions, caregiving responsibilities, health, and social support. These findings support the need for a women-focused analysis of life satisfaction using recent Canadian population-level data.

2.3 Socioeconomic Determinants of Life Satisfaction

Socioeconomic conditions are among the most important determinants of life satisfaction and subjective well-being. Income, education, financial security, and material resources shape individuals’ ability to meet basic needs, access services, manage stress, and maintain quality of life. For women, socioeconomic conditions may be especially important because women are more likely to experience income inequality, caregiving-related work interruptions, and economic vulnerability.

2.3.1 Income and Life Satisfaction

Income is strongly associated with life satisfaction because it affects access to material resources, housing, healthcare, food, transportation, and other necessities. Financial security can reduce stress and provide greater control over life circumstances.

Kahneman and Deaton (2010) distinguished between emotional well-being and life evaluation. Their study found that income was more strongly associated with life evaluation than with day-to-day emotional well-being. They argued that higher income improves how individuals evaluate their lives overall, even though emotional well-being may not continue to improve beyond a certain income level. This distinction is directly relevant to the current study because life satisfaction is closer to life evaluation than to a clinical measure of mental health.

The relationship between income and life satisfaction is important for understanding well-being among women in Canada. Lower-income women may face greater financial strain, difficulty meeting household needs, housing insecurity, and reduced access to resources. These pressures may reduce overall life satisfaction. Therefore, household income is included in this study as a key socioeconomic predictor.

2.3.2 Food Security and Life Satisfaction

Food security is another important socioeconomic determinant of well-being. Household food insecurity reflects inadequate or insecure access to food due to financial constraints. It is not only a food-related issue, but also a broader indicator of material deprivation and economic hardship.

Tarasuk et al. (2023) reported that household food insecurity in Canada reached a record high in 2022, with 17.8% of households in the ten provinces experiencing some level of food insecurity. The report emphasized that food insecurity is strongly linked to low income, limited assets, debt, and other forms of socioeconomic disadvantage. It also identified food insecurity as a major social determinant of health, associated with poorer physical and mental health, greater healthcare use, hospitalization, and premature mortality.

Food insecurity is particularly relevant to women’s well-being because the report found high vulnerability among female lone-parent households and households relying on social assistance or Employment Insurance. These findings suggest that food insecurity may reflect broader financial instability and material hardship that can negatively affect life satisfaction.

Because food security emerged as one of the strongest factors in the exploratory analysis, it is an important variable in the current study. Examining food security allows this research to investigate how material deprivation is associated with subjective well-being among women in Canada.

2.4 Employment and Life Satisfaction

Employment is an important determinant of life satisfaction because it affects income, social participation, identity, daily routine, and access to resources. Employment can contribute positively to well-being, but the relationship depends on the nature and quality of work. Unemployment, unstable employment, or poor-quality employment can negatively affect well-being.

Clark and Oswald (1994) examined the relationship between unemployment and well-being using data from the British Household Panel Study. Their findings showed that unemployed individuals had substantially higher levels of psychological distress than employed individuals. The study concluded that unemployment has a strong negative association with well-being, suggesting that work provides benefits beyond income, including social connection, purpose, and structure.

Employment quality is also important. Butterworth et al. (2011) argued that employment

does not automatically improve mental health or well-being if working conditions are poor. Their study found that low-quality jobs characterized by insecurity, low control, and high stress may negatively affect mental well-being. This suggests that both employment status and employment conditions should be considered when examining well-being.

Women may experience employment-related pressures differently because of caregiving responsibilities, work-family conflict, occupational segregation, and unequal domestic labour. Bilodeau et al. (2025) found that work-family conflict contributes to gendered mental health inequalities across Canadian provinces. These findings suggest that employment and family responsibilities can jointly shape women’s well-being.

In the current study, employment-related variables are included to examine how labour force participation and work status are associated with life satisfaction among women. This is important because employment may influence life satisfaction both directly and indirectly through income, social engagement, and economic stability.

2.5 Health and Life Satisfaction

Health is closely connected to life satisfaction because physical functioning, chronic conditions, and limitations in daily activities can affect independence, social participation, employment, and quality of life. Individuals with poorer health often face barriers that reduce their ability to participate fully in work, family, and community life.

Diener (1984) identified health as one of the factors associated with subjective well-being. Health can influence life satisfaction directly through pain, mobility, and physical limitations, and indirectly through its effects on employment, income, social participation, and daily functioning.

Functional difficulty is especially important because it reflects limitations in performing daily activities. Women experiencing functional difficulties may face reduced independence, increased caregiving needs, lower employment participation, and higher stress. These conditions may reduce life satisfaction more directly than the presence of a chronic condition alone.

Musculoskeletal chronic conditions may also affect well-being because they can contribute to pain, reduced mobility, physical limitations, and difficulty completing everyday activities. Therefore, this study includes both functional difficulty and musculoskeletal chronic conditions as health-related predictors of life satisfaction.

2.6 Post-Pandemic Context and Women’s Well-Being

The COVID-19 pandemic intensified many social and economic pressures affecting women’s well-being. Although this study does not directly measure pandemic experiences, the use of CCHS 2022 data makes the post-pandemic context important. The pandemic affected employment, caregiving responsibilities, household income, food security, social relationships, and health service access.

Park (2024) found that women and girls from diverse backgrounds in Canada experienced different mental health impacts before and during the COVID-19 pandemic. The study emphasized that women facing multiple forms of disadvantage, including low income, disability, unemployment, and social marginalization, reported poorer well-being outcomes. These findings support the importance of examining women’s well-being through a social determinants and intersectional lens.

The pandemic also highlighted the importance of financial security and household resources. Rising costs of living, employment disruptions, and caregiving pressures may have affected women’s life satisfaction beyond the immediate pandemic period. Therefore, recent post-pandemic survey data provide an important opportunity to examine how socioeconomic, employment-related, and health-related factors are associated with life satisfaction among women in Canada.

2.7 Interpretable Modeling in Well-Being Research

Data-driven methods are increasingly used in public health and social science research to examine patterns in large survey datasets. Predictive models can help identify which variables are most strongly associated with well-being outcomes. However, in public health research, interpretability is as important as predictive accuracy because researchers need to understand why certain groups may experience lower well-being.

This study uses interpretable modeling approaches, including Logistic Regression and Decision Tree Classification. Logistic Regression is useful because it provides interpretable estimates of the relationship between predictors and the outcome. Decision Trees are useful because they can identify non-linear patterns and provide visual decision rules.

The purpose of modeling in this study is not only to predict lower life satisfaction, but also to identify which socioeconomic, employment-related, demographic, and health-related factors are most strongly associated with life satisfaction among women. This approach supports both statistical interpretation and practical understanding.

2.8 Research Gaps

Existing literature provides important insights into subjective well-being, income, employment, food security, health, and women’s well-being. However, several gaps remain.

First, many studies examine life satisfaction in general populations, but fewer focus specifically on women using recent Canadian national survey data. Women’s well-being may be shaped by distinct social and economic pressures, including caregiving responsibilities, income inequality, work-family conflict, and health-related limitations.

Second, many studies examine single determinants of well-being, such as income, employment, or health, separately. Fewer studies examine how multiple domains, including socioeconomic status, food security, employment, demographic characteristics, and health-related factors, are jointly associated with life satisfaction among women.

Third, food security is often studied as a health or poverty issue, but it is less often integrated into predictive models of life satisfaction. Given the strong relationship between food insecurity, material deprivation, and poor health outcomes, food security may be an important determinant of subjective well-being.

Fourth, much of the existing literature focuses on either clinical mental health outcomes or general well-being measures. The current study contributes by focusing on life satisfaction as a subjective well-being outcome, which is especially useful when clinical mental health variables are not available for the full population sample.

Finally, although machine learning and predictive methods are increasingly used in health research, many studies prioritize accuracy over interpretation. There remains a need for interpretable data-driven approaches that identify meaningful predictors of well-being in a way that can be understood in public health and social science contexts.

The current study addresses these gaps by using the CCHS 2022 dataset to examine factors associated with life satisfaction among women in Canada. It integrates demographic, socioeconomic, employment-related, and health-related variables within a single analytical framework and applies interpretable modeling methods to identify factors associated with lower life satisfaction.

2.9 Summary of Literature Review

The reviewed literature shows that life satisfaction is a valid and widely used indicator of subjective well-being. Foundational work on subjective well-being emphasizes that life

satisfaction reflects individuals’ overall evaluation of their lives and is influenced by both personal and social conditions.

The literature also shows that women’s life satisfaction is shaped by multiple factors, including socioeconomic conditions, employment experiences, health status, social support, and gendered responsibilities. Income and food security are particularly important because they reflect access to material resources and financial stability. Employment may contribute to well-being through income, structure, identity, and social participation, while unemployment or poor-quality employment may reduce well-being. Health-related factors, including functional difficulties and chronic conditions, may also reduce life satisfaction by limiting independence and daily functioning.

Overall, this literature review supports the need for an integrated analysis of life satisfaction among women in Canada. The current study builds on existing research by using recent CCHS 2022 data to examine how demographic, socioeconomic, employment-related, and health-related factors are associated with life satisfaction. By applying interpretable methods such as Logistic Regression and Decision Tree Classification, the study aims to identify meaningful patterns and provide practical insights into factors associated with women’s subjective well-being in Canada.

3 Data Description

3.1 Dataset Overview

This study uses data from the 2022 Canadian Community Health Survey (CCHS), a national cross-sectional survey conducted by Statistics Canada. The CCHS collects information on health status, health determinants, and healthcare utilization among Canadians aged 12 years and older living in the ten provinces and three territories. The survey is designed to provide reliable health-related information at both national and provincial levels and is widely used for population health research in Canada.

The CCHS contains a broad range of demographic, socioeconomic, employment, and health-related variables, making it a valuable resource for examining factors associated with well-being and health outcomes. The 2022 cycle is particularly relevant because it provides recent post-pandemic information on Canadian populations and captures conditions that may continue to influence health and well-being following the COVID-19 pandemic.

For this study, a subset of variables related to demographic characteristics, socioeconomic status, employment conditions, health status, and life satisfaction was selected. These variables

were chosen based on findings from the literature review and their relevance to understanding well-being among women in Canada.

Table 1: Summary of the CCHS 2022 Dataset Used in This Study

Characteristic	Description
Dataset	Canadian Community Health Survey (CCHS) 2022
Data Source	Statistics Canada
Study Design	Cross-sectional survey
Target Population	Canadians aged 12 years and older
Study Population	Female respondents only
Final EDA Sample Size	52,884 respondents
Geographic Coverage	Canadian provinces and territories
Outcome Variable	Life Satisfaction (LSMDVSWL)
Key Predictor Domains	Demographic, Socioeconomic, Employment, and Health Factors

3.2 Study Population

The analysis focuses exclusively on female respondents within the CCHS 2022 dataset. After filtering the dataset to include women with valid responses for life satisfaction and selected demographic, socioeconomic, and health-related variables, the final exploratory analysis sample consisted of 52,884 respondents.

Restricting the analysis to women allows for a more focused examination of factors associated with well-being within a population that has been shown in previous research to experience unique occupational, socioeconomic, and caregiving-related pressures. This approach is consistent with the objectives of the study, which aims to understand how employment-related, socioeconomic, and health-related factors relate to life satisfaction among women in Canada.

The sample includes women from different age groups, educational backgrounds, income levels, employment situations, and geographic regions across Canada. This diversity allows the study to explore how multiple factors are associated with differences in life satisfaction and overall well-being.

3.3 Variables of Interest

The primary outcome variable used in this study is *Life Satisfaction* (LSMDVSWL). Statistics Canada derives this variable from respondents' self-reported satisfaction with life and groups responses into five categories: Very Satisfied, Satisfied, Neither Satisfied nor Dissatisfied, Dissatisfied, and Very Dissatisfied.

Life satisfaction was selected as the outcome variable because it is available for the majority of respondents and represents a widely used measure of subjective well-being. While it is not a clinical measure of mental illness, life satisfaction captures important aspects of psychological well-being and overall quality of life, making it suitable for examining factors associated with well-being among women in Canada.

During exploratory analysis, life satisfaction is examined using its original five-category structure. For the later predictive modeling stage, this variable may also be transformed into a binary outcome representing higher and lower levels of life satisfaction. In that case, respondents classified as Dissatisfied or Very Dissatisfied will be considered as experiencing lower life satisfaction, while respondents classified as Satisfied or Very Satisfied will be considered as having higher life satisfaction. Neutral responses may be examined separately or handled according to the final modeling strategy.

Several independent variables were selected based on their relevance to the research objectives and findings from prior literature. These variables are grouped into four categories:

- **Demographic Variables**

- Age Group (DHHGAGE)
- Household Size (DHHDGHSZ)
- Province of Residence (GEOGPRV)
- Immigrant Status (SDCDGIMM)
- Visible Minority Status (SDCDVFLA)

- **Socioeconomic Variables**

- Household Income (INCDGHH)
- Education Level (EDDVH3)
- Household Food Security (FSCDVHF2)

- **Employment Variables**

- Labour Force Status / Working Status Last Week (LBFDGWSS)
- Full-Time or Part-Time Employment (LBFDVPFT)
- Hours Worked per Week (LBFDGHPW)

- **Health-Related Variables**

- Difficulty Across Functional Domains (WDMDGDIF)
- Musculoskeletal Chronic Condition (CCCDGSKL)

The selected variables were chosen based on findings from the literature review and their availability within the CCHS 2022 dataset. Previous research has identified income, education, employment conditions, food security, physical health, functional difficulty, and demographic characteristics as important determinants of well-being and mental health-related outcomes. Including variables from multiple domains allows the study to examine the combined influence of socioeconomic, employment-related, demographic, and health-related factors on life satisfaction among women.

3.4 Dataset Limitations

Several limitations should be considered when interpreting the results of this study. First, the CCHS is a cross-sectional survey, meaning that information is collected at a single point in time. Consequently, the analysis can identify associations between variables but cannot establish causal relationships.

Second, many variables within the survey are self-reported and may therefore be subject to recall bias or response bias. Respondents may interpret questions differently or provide socially desirable responses, which may affect the accuracy of some measures.

Third, some employment-related variables contain missing or non-applicable responses because not all respondents were employed at the time of the survey. For example, variables related to hours worked and full-time or part-time employment are primarily applicable to respondents who were working. Therefore, these variables are interpreted carefully during exploratory analysis and may require separate treatment during the modeling stage.

Fourth, life satisfaction represents a subjective measure of well-being rather than a clinical diagnosis of a mental health condition. Although a depression-related variable was explored, it was available only for a limited subset of respondents and was therefore not suitable as the primary outcome variable for the full analysis. Life satisfaction was selected because it was available for the majority of respondents and provides a meaningful measure of subjective well-being and quality of life.

Despite these limitations, the CCHS remains one of the most comprehensive and widely used population health datasets in Canada and provides a strong foundation for examining factors associated with well-being among women in the post-pandemic Canadian context.

4 Exploratory Data Analysis

4.1 Data Cleaning and Preprocessing

The exploratory data analysis (EDA) phase was conducted to understand the characteristics of the study population and to examine the relationships between life satisfaction and selected employment, socioeconomic, and health-related factors. Data cleaning involved removing records containing invalid or non-response categories for the primary outcome variable and key explanatory variables. Specifically, respondents with “Not Stated” or missing responses for life satisfaction, household income, education, food security, functional difficulty, chronic condition status, immigrant status, and visible minority status were excluded from the analytical sample.

During the initial stages of data preparation, a highly restrictive filtering strategy was applied to employment-related variables, resulting in a dataset that primarily contained women who were actively employed. Exploratory analysis revealed that this approach substantially reduced the representativeness of the sample by excluding women who were unemployed, retired, students, homemakers, or otherwise outside the labour force. Consequently, a broader EDA dataset was constructed to retain valid observations for the outcome variable while preserving variation in employment status and other explanatory factors. This approach resulted in a final analytical sample of 52,884 women and provided a more representative basis for exploratory analysis.

4.2 Characteristics of the Study Population

The final analytical sample consisted of 52,884 women from the Canadian Community Health Survey (CCHS) 2022. The study population was primarily composed of women aged between 18 and 64 years. Approximately 36.9% of respondents were between 35 and 49 years of age, while 36.4% were between 50 and 64 years old. Women aged 18 to 34 years represented 26.7% of the sample.

Educational attainment was generally high, with approximately 74.5% of respondents reporting post-secondary education, 18.5% reporting secondary school graduation, and 7.0% reporting less than secondary education. Household income levels were relatively diverse, although 56.0% of respondents belonged to the highest household income category. The remaining respondents were distributed across lower and middle-income categories.

Most respondents (85.7%) reported being food secure. However, approximately 14.3% experienced some degree of food insecurity, including marginal, moderate, or severe food inse-

curity. Employment status also varied across respondents, with 47.3% reporting that they had worked during the previous week, 14.0% reporting that they had not worked, and 38.5% belonging to categories where employment-related questions were not applicable.

These descriptive statistics indicate substantial variation in employment, socioeconomic, and health-related characteristics, providing an appropriate basis for investigating factors associated with life satisfaction.

4.3 Distribution of Life Satisfaction

Life satisfaction was measured using the grouped variable LSMDVSWL, which categorizes respondents into five levels ranging from ‘Very Dissatisfied’ to ‘Very Satisfied.’ Figure 1 presents the distribution of life satisfaction among women in the analytical sample.

The results indicate that the majority of women reported positive levels of life satisfaction. Approximately 30.1% of respondents were classified as very satisfied, while 55.8% reported being satisfied. Smaller proportions reported neutral satisfaction (7.8%), dissatisfaction (5.2%), and very low satisfaction (1.1%). Overall, more than 85% of respondents reported being either satisfied or very satisfied with their lives, suggesting generally favourable levels of subjective well-being within the study population.

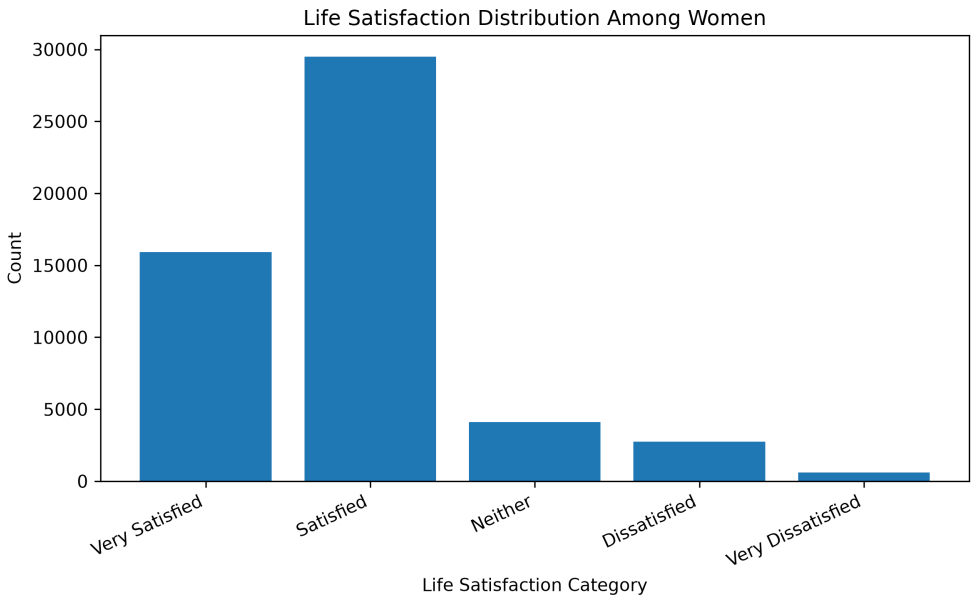


Figure 1: Distribution of Life Satisfaction Among Women

4.4 Employment and Life Satisfaction

Employment status was examined to assess its relationship with life satisfaction. Figure 2 presents the percentage of women reporting low life satisfaction across employment categories.

Women who reported working during the previous week exhibited lower levels of dissatisfaction compared to women who had not worked. Among women who worked during the previous week, approximately 6.0% reported low life satisfaction. In contrast, women who did not work during the previous week reported substantially higher levels of dissatisfaction, with approximately 11.5% reporting low life satisfaction.

Women outside the labour force demonstrated intermediate levels of dissatisfaction. These findings suggest that labour force participation may be associated with improved subjective well-being. Potential explanations include greater financial security, social engagement, and routine associated with employment. While causality cannot be inferred from the cross-sectional data, the results indicate a meaningful relationship between employment status and life satisfaction.

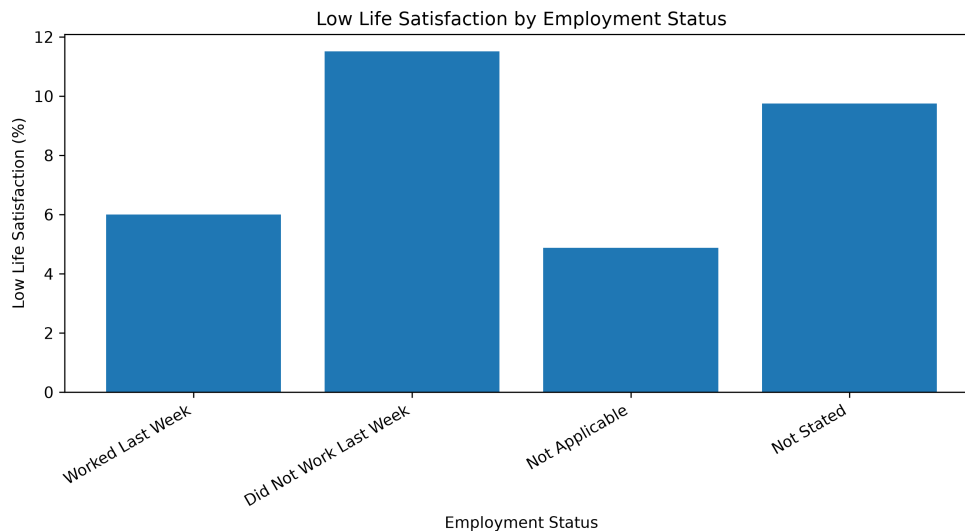


Figure 2: Low Life Satisfaction by Employment Status

4.5 Socioeconomic Factors and Life Satisfaction

4.5.1 Household Income

Household income exhibited a clear relationship with life satisfaction. Figure 3 illustrates the prevalence of low life satisfaction across income groups.

A strong socioeconomic gradient was observed. Women in the lowest income category reported substantially higher levels of dissatisfaction than women in the highest income category. Approximately 16.8% of women in the lowest income group reported low life satisfaction compared with only 4.9% among women in the highest income group. Women in the lowest income category were therefore more than three times as likely to report low life satisfaction as those in the highest income category.

These findings suggest that economic resources play an important role in shaping subjective well-being and are consistent with previous literature linking financial security to improved quality of life.

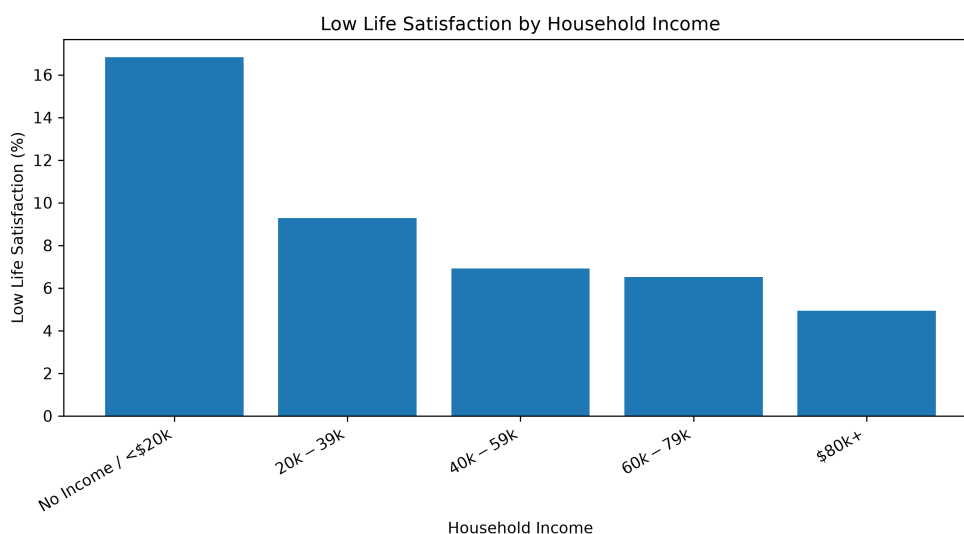


Figure 3: Low Life Satisfaction by Household Income

4.5.2 Food Security

Food security demonstrated the strongest association with life satisfaction among all variables examined in this study. Figure 4 presents the prevalence of low life satisfaction across levels of food security.

A clear and progressive increase in dissatisfaction was observed as food insecurity worsened. Among food-secure women, approximately 4.7% reported low life satisfaction. This increased to 8.9% among marginally food insecure respondents, 14.0% among moderately food insecure respondents, and 28.3% among severely food insecure respondents.

The results suggest that women experiencing severe food insecurity are approximately six times more likely to report low life satisfaction than food-secure women. These findings high-

light the substantial influence of material hardship and economic vulnerability on subjective well-being.

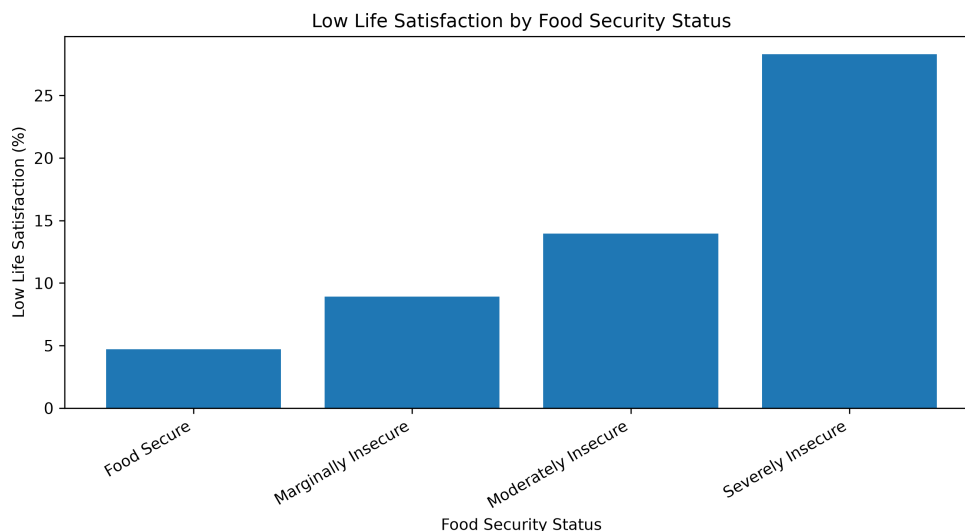


Figure 4: Low Life Satisfaction by Food Security Status

4.5.3 Education

Educational attainment showed a comparatively weaker relationship with life satisfaction. Although women with post-secondary education reported slightly higher levels of satisfaction than women with lower educational attainment, the differences between educational categories were relatively modest.

Compared with income and food security, education appeared to have a weaker association with subjective well-being in this sample. This suggests that current socioeconomic conditions may have a stronger influence on life satisfaction than educational attainment alone.

4.6 Health Factors and Life Satisfaction

4.6.1 Functional Difficulty

Functional difficulty emerged as one of the strongest health-related correlates of life satisfaction. Figure 5 presents the prevalence of low life satisfaction among women with and without functional difficulties.

Women reporting functional difficulties experienced substantially higher levels of low life satisfaction than women without difficulties. Approximately 9.9% of women with functional

difficulties reported low life satisfaction, compared with only 3.0% among women without difficulties.

Women reporting functional difficulties were therefore more than three times as likely to experience low life satisfaction. This finding highlights the importance of physical and functional well-being in shaping overall quality of life.

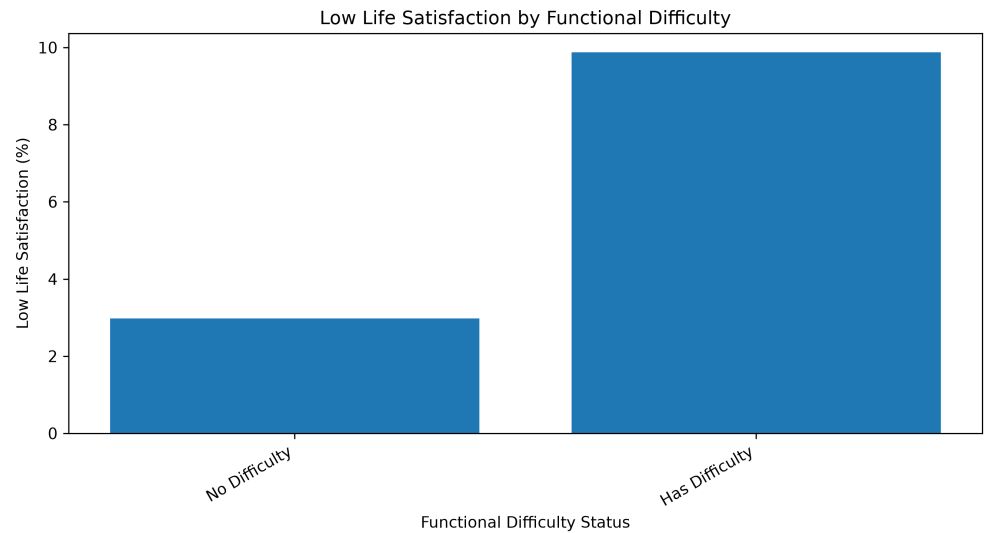


Figure 5: Low Life Satisfaction by Functional Difficulty

4.6.2 Musculoskeletal Chronic Conditions

Musculoskeletal chronic conditions, including arthritis, osteoporosis, and fibromyalgia, were also associated with life satisfaction. Women reporting these conditions were more likely to report low life satisfaction than women without such conditions.

Approximately 8.2% of women with musculoskeletal chronic conditions reported low life satisfaction, compared with 5.5% among women without chronic conditions. Although this relationship was weaker than that observed for functional difficulty, it nevertheless suggests that chronic health conditions may negatively affect subjective well-being.

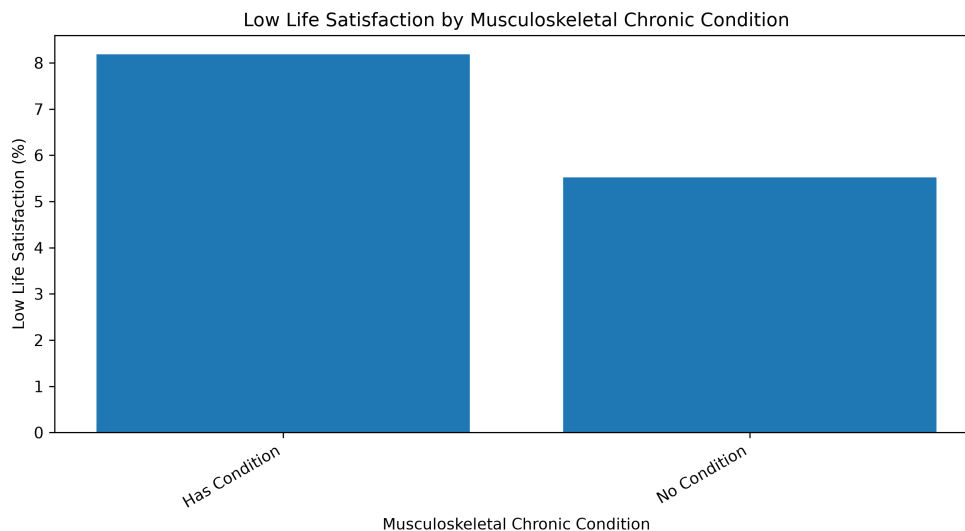


Figure 6: Low Life Satisfaction by Musculoskeletal Chronic Conditions

4.7 Geographic Variation in Life Satisfaction

Geographic variation in life satisfaction was examined using province-level indicators. Differences in the prevalence of low life satisfaction were observed across provinces; however, these differences were relatively modest compared with those observed for socioeconomic and health-related factors.

The prevalence of low life satisfaction ranged from approximately 4% to 8% across provinces. While regional variation may reflect differences in local economic conditions, access to services, and demographic composition, province appeared to be a less influential factor than food security, income, employment status, and functional difficulty. Consequently, province was retained as a contextual variable rather than a primary explanatory factor.

4.8 Summary of Key Findings

Several important findings emerged from the exploratory analysis. First, food security demonstrated the strongest association with life satisfaction, with severely food insecure women reporting substantially higher levels of dissatisfaction than food-secure women. Second, household income exhibited a clear socioeconomic gradient, with lower-income women experiencing markedly lower levels of life satisfaction. Third, functional difficulties were strongly associated with reduced well-being, suggesting that physical functioning plays an important role in overall quality of life.

Employment status also demonstrated a meaningful relationship with life satisfaction, with women who were not working reporting higher levels of dissatisfaction than women who were employed. Musculoskeletal chronic conditions showed moderate associations with life satisfaction, while education and province exhibited comparatively weaker relationships.

Overall, the exploratory analysis suggests that life satisfaction among women in Canada is shaped by a combination of socioeconomic, employment-related, and health-related factors. Among the variables examined, food security, household income, and functional difficulty emerged as the strongest correlates of subjective well-being. These findings provide a strong empirical foundation for the predictive modeling approaches presented in the next chapter.

5 Project Approach

5.1 Research Methodology

This study uses a quantitative secondary-data research design based on the 2022 Canadian Community Health Survey (CCHS). Since the CCHS is a cross-sectional survey, the purpose of the analysis is to identify associations and predictive patterns rather than establish causal relationships. The study focuses on female respondents and examines how demographic, socioeconomic, employment-related, and health-related factors are associated with life satisfaction and subjective well-being.

The research methodology follows a structured data analysis pipeline. First, relevant variables were selected from the CCHS 2022 dataset based on the literature review and the objectives of the study. Second, data cleaning and preprocessing were conducted to remove invalid or non-response categories and prepare the dataset for analysis. Third, exploratory data analysis was performed to understand the distribution of life satisfaction and examine bivariate relationships between life satisfaction and key explanatory variables. Fourth, supervised learning models will be developed to examine whether selected variables can predict lower life satisfaction among women.

Two modeling approaches will be used: Logistic Regression and Decision Tree Classification. Logistic Regression is included because it provides interpretable estimates of the association between predictors and the outcome. Decision Tree Classification is included because it can identify non-linear patterns and interaction-like structures among variables. Together, these methods support both statistical interpretation and practical understanding of the factors associated with life satisfaction.

This approach is appropriate for the current study because the objective is not only to

predict life satisfaction, but also to understand which socioeconomic, employment-related, and health-related factors appear most relevant. Therefore, the methodology emphasizes interpretability, transparency, and connection to public health and social science research.

5.2 Data Preparation and Feature Selection

The initial dataset was obtained from the 2022 Canadian Community Health Survey. Female respondents were selected to align with the objectives of the study. Variables were chosen based on findings from the literature review and their availability within the survey dataset.

The selected variables were grouped into four domains:

- **Demographic variables:** age group, household size, province of residence, immigrant status, and visible minority status.
- **Socioeconomic variables:** household income, educational attainment, and household food security.
- **Employment-related variables:** labour force status, hours worked, and full-time or part-time employment.
- **Health-related variables:** functional difficulty and musculoskeletal chronic conditions.

Data cleaning procedures involved removing invalid, non-response, and not-stated values for key variables. During exploratory analysis, it was determined that employment-related variables required special consideration because some questions were only applicable to respondents who were employed at the time of the survey. Therefore, a broader exploratory analysis dataset was created to preserve variation in employment status and improve representativeness.

The final exploratory analysis dataset consisted of 52,884 women with valid responses for life satisfaction and selected explanatory variables.

5.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted before model development to understand the structure of the dataset and identify potential relationships between explanatory variables and life satisfaction.

The EDA process included:

- Descriptive analysis of demographic, socioeconomic, employment, and health-related characteristics.
- Examination of the distribution of life satisfaction.
- Cross-tabulation analysis between life satisfaction and selected explanatory variables.
- Visualization of relationships using charts and comparative plots.
- Identification of variables showing strong associations with life satisfaction.

The exploratory analysis showed that food security, household income, functional difficulty, employment status, and musculoskeletal chronic conditions were meaningfully associated with life satisfaction. These findings support the inclusion of these variables in the later predictive modeling stage.

5.4 Predictive Modeling Approach

Following exploratory analysis, supervised learning techniques will be applied to examine whether selected variables can predict lower life satisfaction among women.

The outcome variable will be derived from the grouped life satisfaction measure (LSMDVSWL). For classification purposes, respondents reporting Dissatisfied or Very Dissatisfied may be classified as experiencing lower life satisfaction, while respondents reporting Satisfied or Very Satisfied may be classified as experiencing higher life satisfaction. Neutral responses will be examined carefully during model development and handled according to the final modeling strategy.

Two complementary predictive modeling approaches will be used: Logistic Regression and Decision Tree Classification.

5.4.1 Logistic Regression

Logistic Regression is a widely used statistical learning method for binary classification problems and is commonly applied in public health and social science research. The method estimates the probability of an outcome occurring based on a set of explanatory variables.

Logistic Regression was selected for this study because it provides interpretable model coefficients and allows the direction and strength of associations between predictors and the outcome to be examined. This is important because the study aims not only to predict lower life satisfaction, but also to understand which factors are most strongly associated with it.

5.4.2 Decision Tree Classification

Decision Tree Classification is a non-parametric machine learning technique that recursively partitions the data into subgroups based on predictor variables. Unlike Logistic Regression, Decision Trees can capture non-linear relationships and interaction-like patterns among predictors.

Decision Trees were selected because they provide intuitive and interpretable decision rules. This makes them useful for identifying higher-risk profiles of women based on combinations of socioeconomic, employment-related, and health-related factors. The Decision Tree model will complement the Logistic Regression model by providing a more visual and rule-based understanding of the data.

5.5 Model Evaluation

Model performance will be evaluated using a train-test validation framework. The analytical dataset will be divided into training and testing subsets to assess how well the models generalize to unseen data.

Several classification metrics will be used to evaluate model performance:

- Accuracy
- Precision
- Recall
- F1-score
- Receiver Operating Characteristic Area Under the Curve (ROC-AUC)

Because lower life satisfaction represents a smaller group within the dataset, evaluation will not rely on accuracy alone. Precision, recall, F1-score, and ROC-AUC will also be considered to better assess model performance for identifying respondents with lower life satisfaction.

5.6 Project Workflow

The overall workflow of the study consists of data acquisition, variable selection, data cleaning, exploratory analysis, feature engineering, predictive modeling, model evaluation, and interpretation. The workflow is illustrated in Figure 7.

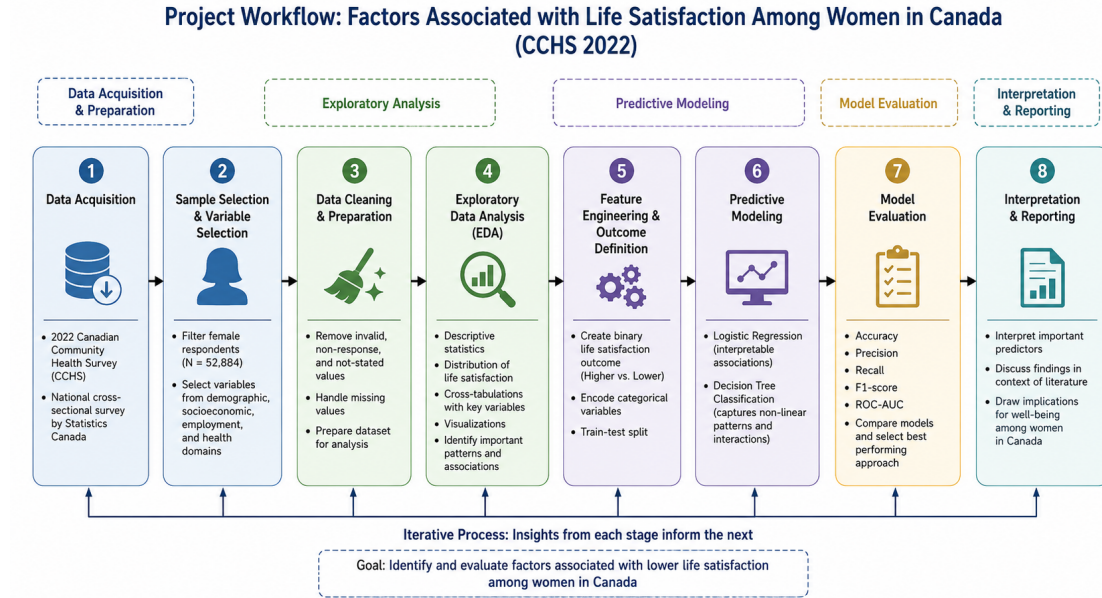


Figure 7: Overall Project Methodology

This workflow provides a systematic and reproducible approach for examining factors associated with life satisfaction among women in Canada. The results from the exploratory analysis and predictive modeling stages will be used to identify the most important determinants of lower life satisfaction and to support interpretation within a public health and social science context.

6 Project Repository

The source code, notebooks, figures, and project documentation associated with this study are available through the following GitHub repository:

https://github.com/Khushboosingh1988/MRP_CCHS_2022

The repository contains the data preparation workflow, exploratory data analysis notebooks, project documentation, figures, and supporting materials used throughout the study. Due to Statistics Canada licensing restrictions, the CCHS 2022 dataset is not included in the repository.

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